

The First-Order Contrast Operator

The elastodynamic matrix–vector wave equation as a route to the single-site T -matrix: derivation, validation, and what it actually buys

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Abstract

The elastodynamic matrix–vector wave equation $\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q} + \mathbf{d}$ expresses the wave field as a first-order recursion in depth. This note asks whether the cube moment method can be applied to it directly, to obtain a single-site T -matrix without passing through the Navier equation. The answer is yes, and the construction is not new: the thesis already carries it in 2 $\frac{1}{2}$ -D. What is new here is the 3-D Cartesian operator, and a measurement of what the route is worth.

Three results. *First*, the operator matrix $\mathcal{A}$ is derived from the stiffness tensor and validated against two independent objects that agree under one constant similarity transform; the thesis shorthand ζ, χ are shown to be the paper’s v_1, v_2 . *Second*, the test space is *derived* rather than chosen: $\mathbf{J}_6$ is antisymmetric and $\Delta \mathcal{A}^t \mathbf{J}_6 = -\mathbf{J}_6 \Delta \mathcal{A}$, so $\mathbf{J}_6 \Delta \mathcal{A}$ is symmetric and the only natural projection is $\text{test} = \mathbf{J}_6 \times \text{trial}$. This dissolves an apparent mismatch of ten conditions against nine unknowns. *Third*, and least expected, the resulting coupling agrees with the ordinary Navier weak form at Born order but differs at $O(\Delta c^2)$ entirely within the traction channel — and that difference splits into a genuine O_h -invariant resummation *and* a spurious preference for the x_3 axis, in comparable amounts. A single-site score against an independent arbiter in a homogeneous background would mix the two and settle neither. The directional preference ceases to be an artefact once the background is layered, which is where the route was wanted.

The T -matrix itself is assembled in §9 and the comparison is reported channel by channel. The sharpest statement is not a norm: e_{12} and e_{13} are equivalent under O_h and so must amplify identically, and in the first-order T they split by 8.4×10^{-4} while the ordinary route splits them by exactly zero. That split is an arbiter-free *lower bound* on the first-order error, $\geq \delta/2$; it bounds nothing about the ordinary route, which can be symmetrically wrong, so no ordering of the two follows from it.

Every claim below is backed by an executable check; the scripts report 25/25, 44/44 and 12/12, and a separate gate binds the numerical propagator to the derived $\mathcal{A}$. Section 12 states plainly what is *not* yet established.

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1 Conventions and objects

1.1 Indices, time, and units

The positive x_3 axis points downward; x_1, x_2 are lateral. Greek indices α, β run over the lateral pair $\{1, 2\}$ only, Latin indices over $\{1, 2, 3\}$. The time convention is $e^{-i\omega t}$ throughout, so $\partial_t \rightarrow -i\omega$

and the particle velocity is $\mathbf{v} = -i\omega\mathbf{u}$. The lateral transform is $e^{+i(k_1x_1+k_2x_2)}$, hence $\partial_\alpha \rightarrow ik_\alpha$.

Numerical work uses seismic units: km, km s^{-1} , g cm^{-3} , GPa. These are consistent, since $\text{GPa} = \text{g cm}^{-3} \cdot (\text{km s}^{-1})^2$.

Two orderings appear in the sources and must not be conflated. The paper orders components (1, 2, 3); the thesis orders them $(z, x, y) = (x_3, x_1, x_2)$. The map between them is written explicitly in §3.2.

1.2 The field vector

The matrix–vector wave equation is

$$\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q} + \mathbf{d}, \quad \mathbf{q} = \begin{pmatrix} \mathbf{q}_1 \\ \mathbf{q}_2 \end{pmatrix}, \quad \mathcal{A} = \begin{pmatrix} \mathcal{A}_{11} & \mathcal{A}_{12} \\ \mathcal{A}_{21} & \mathcal{A}_{22} \end{pmatrix}, \quad (1)$$

with the elastodynamic choice

$$\mathbf{q}_1 = -\boldsymbol{\tau}_3, \quad \mathbf{q}_2 = \mathbf{v}, \quad (\boldsymbol{\tau}_j)_i = \tau_{ij}, \quad (2)$$

so that $\mathbf{q} = (-\tau_{13}, -\tau_{23}, -\tau_{33}, v_1, v_2, v_3)^t$. This is not an arbitrary pairing: $-\boldsymbol{\tau}_3$ and $\mathbf{v}$ are conjugate in the vertical power flux, $j = \frac{1}{4}(-\boldsymbol{\tau}_3^\dagger \mathbf{v} - \mathbf{v}^\dagger \boldsymbol{\tau}_3)$, and that is what makes the metric of §1.3 the natural one.

1.3 The metric $\mathbf{J}_6$

The symplectic metric is

$$\mathbf{J}_6 = \begin{pmatrix} \mathbf{0} & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0} \end{pmatrix}, \quad \mathbf{J}_6^t = -\mathbf{J}_6, \quad \mathbf{J}_6^2 = -\mathbf{I}_6. \quad (3)$$

A naming warning. This project, following the thesis, calls this matrix $\mathbf{J}_6$; it is already defined as such in the package. The paper calls the *same* matrix $\mathbf{N}$, and uses $\mathbf{J}$ for $\text{diag}(\mathbf{I}_3, -\mathbf{I}_3)$, a different object. Translate on the way in. Every occurrence of $\mathbf{N}$ in the paper’s equations (16)–(30) corresponds to $\mathbf{J}_6$ here.

The operator matrix is *quasi-Hamiltonian* with respect to it,

$$\mathcal{A}^t \mathbf{J}_6 = -\mathbf{J}_6 \mathcal{A}, \quad (4)$$

which is the paper’s eq. (27) and the thesis’s **ATdef**. Everything in §6 follows from (4) together with the antisymmetry of $\mathbf{J}_6$, and from nothing else.

2 The first-order system for an isotropic medium

2.1 From the stiffness tensor to the stiffness matrices

For an isotropic solid,

$$c_{ijkl} = \lambda \delta_{ij} \delta_{kl} + \mu (\delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}), \quad \rho_{ij} = \rho \delta_{ij}, \quad (5)$$

with $\lambda = \lambda(\mathbf{x}, \omega)$, $\mu = \mu(\mathbf{x}, \omega)$, $\rho = \rho(\mathbf{x}, \omega)$ and $K_c = \lambda + 2\mu$. The stiffness matrices are $(\mathbf{C}_{jl})_{ik} = c_{ijkl}$. In particular

$$\mathbf{C}_{33} = \text{diag}(\mu, \mu, K_c), \quad \mathbf{C}_{33}^{-1} = \text{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1}), \quad (6)$$

and $\mathbf{C}_{jl}^t = \mathbf{C}_{lj}$, which follows from $c_{ijkl} = c_{klij}$. All nine blocks are reproduced from (5) and checked entry by entry.

2.2 Eliminating the in-plane tractions

The components $\boldsymbol{\tau}_1, \boldsymbol{\tau}_2$ do not appear in (2) and must be removed. Doing so introduces the reduced stiffness

$$\mathbf{U}_{\alpha l} = \mathbf{C}_{\alpha l} - \mathbf{C}_{\alpha 3} \mathbf{C}_{33}^{-1} \mathbf{C}_{3l}, \quad (7)$$

with the two structural properties $\mathbf{U}_{\alpha 3} = \mathbf{0}$ and $\mathbf{U}_{\alpha\beta}^t = \mathbf{U}_{\beta\alpha}$. Both are checked rather than assumed, because it is they — not the algebra that produced them — that make the symmetry relations of §3.3 hold.

The isotropic $\mathbf{U}_{\alpha\beta}$ is where two constants are *born*:

$$v_1 \equiv (\mathbf{U}_{11})_{11} = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad v_2 \equiv (\mathbf{U}_{12})_{12} = \frac{2\mu\lambda}{\lambda + 2\mu}, \quad v_1 - v_2 = 2\mu. \quad (8)$$

They are read off $\mathbf{U}$ and only then compared with the published definitions; they are not independent quantities. Every $\mathbf{U}_{\alpha\beta}$ has a null third row and column, which is the statement that τ_{33} has been fully eliminated.

2.3 The operator matrices

The general construction gives

$$\mathscr{A}_{11} = -\partial_\alpha \mathbf{C}_{\alpha 3} \mathbf{C}_{33}^{-1}, \quad \mathscr{A}_{12} = i\omega\rho - \frac{1}{i\omega} \partial_\alpha \mathbf{U}_{\alpha\beta} \partial_\beta, \quad \mathscr{A}_{21} = i\omega \mathbf{C}_{33}^{-1}, \quad \mathscr{A}_{22} = -\mathbf{C}_{33}^{-1} \mathbf{C}_{3\beta} \partial_\beta, \quad (9)$$

and for the isotropic medium these evaluate to

$$\mathscr{A}_{11} = \begin{pmatrix} 0 & 0 & -\partial_1 \frac{\lambda}{\lambda+2\mu} \\ 0 & 0 & -\partial_2 \frac{\lambda}{\lambda+2\mu} \\ -\partial_1 & -\partial_2 & 0 \end{pmatrix}, \quad \mathscr{A}_{22} = \begin{pmatrix} 0 & 0 & -\partial_1 \\ 0 & 0 & -\partial_2 \\ -\frac{\lambda}{\lambda+2\mu} \partial_1 & -\frac{\lambda}{\lambda+2\mu} \partial_2 & 0 \end{pmatrix}, \quad (10)$$

$$\mathscr{A}_{12} = \begin{pmatrix} i\omega\rho - \frac{1}{i\omega}(\partial_1 v_1 \partial_1 + \partial_2 v_2 \partial_2) & -\frac{1}{i\omega}(\partial_2 \mu \partial_1 + \partial_1 v_2 \partial_2) & 0 \\ -\frac{1}{i\omega}(\partial_2 v_2 \partial_1 + \partial_1 \mu \partial_2) & i\omega\rho - \frac{1}{i\omega}(\partial_1 \mu \partial_1 + \partial_2 v_1 \partial_2) & 0 \\ 0 & 0 & i\omega\rho \end{pmatrix}, \quad (11)$$

with $\mathcal{A}_{21} = i\omega \operatorname{diag}(\mu^{-1}, \mu^{-1}, K_c^{-1})$.

The ordering is the content. Compare the $(1, 3)$ entry of $\mathcal{A}_{11}$ with the $(3, 1)$ entry of $\mathcal{A}_{22}$. In the first the derivative stands to the *left* of the modulus and differentiates the product; in the second it stands to the *right* and differentiates the field alone. For a laterally invariant medium the distinction collapses and is invisible. For a laterally varying one it is the physics. This is why the checks in §3.1 compare *operators* — by their action on generic fields — rather than expressions, and why they are accompanied by a negative control that corrupts precisely this ordering.

3 Validating $\mathcal{A}$ against two independent objects

3.1 Arbiter A: the published isotropic operator matrices

The blocks (10)–(11) are transcribed separately and compared against those built from (9). The comparison is made by acting on a vector of generic symbolic functions, so it tests the derivative ordering as well as the coefficients. All four match.

A check that cannot fail is worth nothing, so a corrupted $\mathcal{A}_{22}$ — with the derivative wrongly placed outside the modulus — is also tested, and is rejected.

3.2 Arbiter B: the thesis system matrix

The thesis carries a 6×6 Fourier-domain system matrix $\mathbf{A}(k_x, k_y)$ for a laterally invariant medium, in a *different* field vector

$$\mathbf{b} = (u_3, u_1, u_2, \tau_{33}, \tau_{31}, \tau_{32})^t, \quad (12)$$

i.e. displacement–stress rather than velocity–traction, ordered (z, x, y) rather than $(1, 2, 3)$, and normalised so that $\rho\omega^2$ appears where the paper has $i\omega\rho$. It uses the shorthand

$$\gamma = \frac{\lambda}{\lambda + 2\mu}, \quad a = \frac{1}{\lambda + 2\mu}, \quad b = \frac{1}{\mu}, \quad \zeta = \frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad \chi = \frac{2\mu\lambda}{\lambda + 2\mu}. \quad (13)$$

Comparing (8) with these: $\zeta \equiv \nu_1$ and $\chi \equiv \nu_2$. The thesis’s in-plane stiffnesses and the paper’s auxiliary moduli are the same two quantities, born in the same elimination (7). This is verified, not asserted.

Setting λ, μ, ρ constant and acting on $e^{i(k_1 x_1 + k_2 x_2)}$ reduces $\mathcal{A}$ to a matrix. With the constant map $\mathbf{b} = \mathbf{S}\mathbf{q}$ built from $\mathbf{u} = \mathbf{v}/(-i\omega)$, $\boldsymbol{\tau}_3 = -\mathbf{q}_{1:3}$ and the permutation $(1, 2, 3) \rightarrow (3, 1, 2)$, the identity

$$\mathbf{A}_{\text{thesis}} = \mathbf{S} \mathbf{A}_{\text{paper}} \mathbf{S}^{-1} \quad (14)$$

holds entry by entry. Since the two matrices were derived independently and published separately, this is evidence rather than bookkeeping.

3.3 Structural checks

Three further properties are verified on the reduced 6×6 :

1. the operator symmetry relations $\mathcal{A}_{11}^t = -\mathcal{A}_{22}$, $\mathcal{A}_{12}^t = \mathcal{A}_{12}$, $\mathcal{A}_{21}^t = \mathcal{A}_{21}$, where the transpose is the formal adjoint under integration by parts ($\partial_\alpha^t = -\partial_\alpha$);
2. the quasi-Hamiltonian relation (4), which in the wavenumber domain reads $\mathbf{A}^t(-\mathbf{k})\mathbf{J}_6 = -\mathbf{J}_6\mathbf{A}(\mathbf{k})$;
3. the characteristic polynomial factorises as $(s^2 + k_{z,p}^2)(s^2 + k_{z,s}^2)^2$, recovering $\alpha^2 = (\lambda + 2\mu)/\rho$ and $\beta^2 = \mu/\rho$ from a construction in which no wave speed was ever written down.

Item 3 is the one that binds the algebra to physics: nothing upstream knows the wave speeds.

4 The contrast operator

4.1 Definition and the Lippmann–Schwinger equation

Inside the scatterer the medium is $(\lambda + \Delta\lambda, \mu + \Delta\mu, \rho + \Delta\rho)$. Writing $\mathcal{A}^{\text{tot}} = \mathcal{A} + \Delta\mathcal{A}$, the total field solves the background system driven by its own contrast,

$$(\partial_3 - \mathcal{A})\mathbf{q} = \Delta\mathcal{A}\mathbf{q} + \mathbf{d}, \quad (15)$$

which is the Lippmann–Schwinger equation of the first-order system.

4.2 The foundation check

Before (15) means anything, $\mathbf{q}$ must actually satisfy (1). Building $\mathbf{q}$ from a *generic* displacement field in a *laterally varying* medium and forming $\partial_3\mathbf{q} - \mathcal{A}\mathbf{q}$ gives two halves with different strengths:

- rows 4–6 vanish **identically**, for any $\mathbf{u}$ whatever. They are the constitutive relation rearranged; no equation of motion is used. Failure here would mean the elimination (7) is wrong.
- rows 1–3 equal **exactly** the Navier body force $f_i = -\rho\omega^2 u_i - \partial_j\tau_{ij}$. Anything left over would be a defect in $\mathcal{A}_{11}$ or $\mathcal{A}_{12}$.

Both hold. This is a considerably stronger statement than the plane-wave and eigenvalue checks of §3.3, because $\mathbf{u}$ is generic and the moduli vary laterally, so every ordering term in (10) is exercised.

4.3 $\Delta\mathcal{A}$ is not linear in the contrast

The Navier potential is exactly linear in $(\Delta\lambda, \Delta\mu, \Delta\rho)$. $\Delta\mathcal{A}$ is not, because the elimination inverted $\mathbf{C}_{33}$:

$$(\Delta\mathcal{A}_{21})_{11} = i\omega \left(\frac{1}{\mu + \Delta\mu} - \frac{1}{\mu} \right) = -\frac{i\omega \Delta\mu}{\mu(\mu + \Delta\mu)}, \quad (16)$$

whose ratio to its linearisation $-i\omega\Delta\mu/\mu^2$ is exactly $\mu/(\mu + \Delta\mu)$. At the project’s validated moderate contrast ($\alpha = 5$, $\beta = 3$, $\rho = 2.5$, so $\lambda = 17.5$, $\mu = 22.5$ GPa; $\Delta\lambda = 2$, $\Delta\mu = 1$ GPa):

quantity	value
$\Delta\mu/\mu$	0.04444
exact / linear for $\Delta b = \Delta(1/\mu)$	0.95745
exact / linear for $\Delta a = \Delta(1/K_c)$	0.93985

The first-order form has already summed part of the local response exactly. §7 measures what that is worth, and finds the answer is not the obvious one.

5 The augmentation

5.1 Why a derivative must be moved

Inside the scatterer the contrasts are constant, so every coefficient of $\Delta\mathcal{A}$ carries the indicator $\mathbf{1}_V$. A derivative standing to the *left* of such a coefficient differentiates the indicator and produces a surface delta on the cube faces. Integration by parts over all space moves that derivative onto the test function instead, where it meets something smooth:

$$\int \phi_i (-\partial_\alpha (m \mathbf{1}_V \psi_j)) d\mathbf{x} = + \int (\partial_\alpha \phi_i) m \mathbf{1}_V \psi_j d\mathbf{x}, \quad (17)$$

and the face term never arises. This is the step the thesis describes as transferring the derivatives onto the source and receiver coordinates of the propagator, and it is what converts a singular integro-differential equation into a nonsingular, compact integral equation.

5.2 The augmented dimension

After the transfer every term has the form $(\partial^a \phi)^t \cdot \text{const} \cdot (\partial^b \psi)$ with $a, b \in \{0, 1\}$, so the form is *multiplicative* on the enlarged vector $\mathcal{L}\psi = (\psi, \partial_1 \psi, \partial_2 \psi)$ — nominally $6 + 6 + 6 = 18$ slots. The true dimension is smaller and is counted, not assumed.

Result. Exactly $10 = 6 + 4$ slots are used. The four extra are $\partial_1 v_1, \partial_2 v_1, \partial_1 v_2, \partial_2 v_2$: the lateral gradients of the two *horizontal* velocity components, and nothing else.

Setting $\partial_2 \rightarrow ik_y$, as a $2\frac{1}{2}$ -D formulation does, makes two of those four multiplicative, leaving $6 + 2 = 8$. That is precisely the thesis’s count for its augmented coupling matrix, reproduced here from an independent 3-D Cartesian derivation. The agreement is a corroboration of the thesis construction, not a restatement of it.

The transfer identity is verified three ways: the augmented matrix is constant (no derivative operators survive); the augmented form reproduces the transferred bilinear; and the untransferred and transferred bilinears differ by an explicitly exhibited lateral divergence $\partial_\alpha P_\alpha$.

6 The test space, derived

6.1 The argument

At this point there are ten augmented slots and, for an affine trial space, nine parameters. The temptation is to choose nine conditions. That temptation must be resisted: a subset chosen by scoring against the arbiter it is later judged by is a fitted parameter wearing a derivation's clothes.

There is no choice to make. $\mathbf{J}_6$ is *antisymmetric*, and (4) applies to $\Delta\mathcal{A}$ as well as to $\mathcal{A}$ (it is linear in the operator), so

$$(\mathbf{J}_6 \Delta\mathcal{A})^t = \Delta\mathcal{A}^t \mathbf{J}_6^t = \Delta\mathcal{A}^t (-\mathbf{J}_6) = \mathbf{J}_6 \Delta\mathcal{A}. \quad (18)$$

$\mathbf{J}_6 \Delta\mathcal{A}$ is a **symmetric** matrix, so

$$b[\phi, \psi] = \phi^t \mathbf{J}_6 \Delta\mathcal{A} \psi \quad (19)$$

is a symmetric bilinear form. A symmetric form admits exactly one natural projection:

$$\boxed{\text{test space} = \mathbf{J}_6 \times \text{trial space.}} \quad (20)$$

Verified, together with the negative control: drop $\mathbf{J}_6$ and the symmetry is lost, both at operator level and in the projected system.

6.2 The ten-versus-nine mismatch dissolves

The ten slots were never test conditions. They measure how many derivative channels the *form* touches — its representation — not how many independent functionals it supplies. Each trial function ψ_k gives exactly one test functional $(\mathbf{J}_6 \psi_k)^t$, so the number of conditions equals the trial dimension by construction.

6.3 The effective coupling matrix

For the affine trial space — nine parameters, three rigid translations and six uniform strains — the projected coupling

$$(\Delta\mathbf{C}_{\text{eff}})_{kl} = \int_V (\mathcal{L} \psi_k)^t \Delta\mathbf{C} (\mathcal{L} \psi_l) dV \quad (21)$$

is the 3-D Cartesian analogue of the thesis's effective coupling matrix. It is 9×9 , symmetric, of full rank 9, vanishes with the contrast, and — because the trial functions are polynomials — requires **no Green's function at all**: the cube integral is exact.

The rigid-translation block is diagonal,

$$(\Delta\mathbf{C}_{\text{eff}})_{1:3,1:3} = i\omega^3 \Delta\rho V \mathbf{I}_3, \quad V = (2a)^3. \quad (22)$$

The power of ω is a basis statement, not a physical one. $\Delta\mathcal{A}$'s density entry is $i\omega\Delta\rho$ acting on the velocity slot; the trial carries $\mathbf{v} = -i\omega\mathbf{u}$ there; and $\mathbf{J}_6$ pulls the test function's

velocity slot into the traction rows, supplying a second $-i\omega$. Hence $i\omega \cdot (-i\omega) \cdot (-i\omega) = i\omega^3$. The displacement-basis answer $i\omega\Delta\rho V$ is wrong by ω^2 . This is the displacement-versus-flux confusion in a new place, and the check asserts the correct factor *and* asserts the incorrect one is incorrect, so it cannot drift back.

7 What the first-order coupling actually buys

7.1 The trial spaces coincide

It is tempting to attribute any difference from the Navier formulation to the nonlinearity of §4.3, or to a different ansatz. Neither is right. Inside the scatterer the moduli are constant, so $\boldsymbol{\tau}_3 = \mathbf{C}_{33}^{\text{tot}} \boldsymbol{\varepsilon}_{i3}$ is a constant *invertible* map. A uniform strain gives a uniform $\boldsymbol{\tau}_3$ and an affine $\mathbf{v}$, and conversely: “ $\boldsymbol{\varepsilon}$ uniform” and “ $\boldsymbol{\tau}_3$ uniform plus in-plane $\boldsymbol{\varepsilon}$ uniform” describe the same nine-parameter family.

The two formulations are therefore **different projections of the same residual onto the same trial space**. The difference lives in the test conditions.

7.2 The density channel cannot differ

With $\Delta\lambda = \Delta\mu = 0$, $\Delta\mathcal{A}$ is exactly $i\omega\Delta\rho$ in rows 1–3, columns 4–6, and reaches *no* part of the stress-glut slot: a density contrast is a pure body force. The propagator’s body-force velocity block is $-i\omega\mathbf{G}$, so both routes return

$$A_u = \frac{1}{1 - \omega^2 \Delta\rho \Gamma_0}, \quad \Gamma_0 = \int_V \mathbf{G} dV, \quad (23)$$

identically. The modulus channels are the only place the two can part.

7.3 Born agreement, and an $O(\Delta c^2)$ difference

Reading a proportionality constant off the rigid block alone gives $i\omega$, and testing it on the remaining eighty entries **refutes** the hypothesis $\Delta\mathbf{C}_{\text{eff}} = i\omega\Delta\mathbf{C}_{\text{Navier}}$. The two couplings agree exactly at $O(\Delta c^0)$ and $O(\Delta c^1)$ — identical Born limits, so nothing is broken — and differ from second order.

Where. Eleven of eighty-one entries differ. With the parameter order $(c_1, c_2, c_3, e_{11}, e_{22}, e_{33}, e_{12}, e_{13}, e_{23})$ the nonzero block is $\{e_{11}, e_{22}, e_{33}\}^2$ together with (e_{13}, e_{13}) and (e_{23}, e_{23}) .

That is exactly the set of strains $\boldsymbol{\tau}_3$ sees:

$$\tau_{13} = 2\mu e_{13}, \quad \tau_{23} = 2\mu e_{23}, \quad \tau_{33} = \lambda(e_{11} + e_{22} + e_{33}) + 2\mu e_{33}. \quad (24)$$

The in-plane shear e_{12} is **untouched**, as it must be: τ_{12} was eliminated, not resummed. This was a falsifiable prediction about which entries may be nonzero, and it held. A representative entry is

$$(\Delta\mathbf{C}_{\text{eff}} - i\omega\Delta\mathbf{C}_{\text{Navier}})_{e_{11}e_{11}} = -\frac{i\omega V \Delta\lambda^2}{\lambda + 2\mu}, \quad (25)$$

pure second order, divided by K_c : the remainder of the geometric series for $1/(K_c + \Delta K_c)$.

7.4 Half of it is a broken symmetry

An isotropic cube in an isotropic whole space has an O_h -symmetric T -matrix; no direction is special. But the first-order system is a recursion in *depth*: it eliminated τ_1, τ_2 and kept τ_3 , so x_3 is preferred by construction. The tell is already above — (e_{13}, e_{13}) and (e_{23}, e_{23}) differ while (e_{12}, e_{12}) does not, and those three are O_h -equivalent.

Applying the cyclic element $x \rightarrow y \rightarrow z \rightarrow x$ (positive controls: $P^3 = \mathbf{I}$, and the Navier coupling *is* invariant):

contrast (GPa)	invariant	breaking	ratio
pure shear, $\Delta\mu = 1$	2.62×10^{-2}	1.94×10^{-2}	1.3520
moderate, (2, 1, 0.1)	5.17×10^{-4}	2.93×10^{-4}	1.7669
strong, (20, 10, 0.5)	1.05×10^{-2}	5.92×10^{-3}	1.7669

Table 1. Decomposition of $\Delta\mathbf{C}_{\text{eff}} - i\omega\Delta\mathbf{C}_{\text{Navier}}$, relative to $\|\Delta\mathbf{C}_{\text{eff}}\|$, at $\lambda = 17.5$, $\mu = 22.5$ GPa. The two mixed rows share a contrast *direction* ($\Delta\lambda:\Delta\mu = 2:1$) at different magnitudes and return the identical ratio, so the split is a function of contrast direction alone — as a leading-order $O(\Delta c^2)$ split must be.

So the difference is neither pure artefact nor pure physics. It carries both a genuine O_h -invariant resummation the Navier form lacks, and a spurious preference for x_3 .

A caveat on this test. The group used is the cyclic subgroup of order 3, not all 48 elements of O_h . Invariance under it is necessary for O_h invariance, not sufficient.

7.5 What follows, and what does not

Two readings are available and the measurement above does not separate them.

Against the route: an approximation that breaks a symmetry the exact answer possesses has introduced an error that no propagator can remove.

For the route: the correction is not scattered. It lands exactly on the traction channel, treating $\mathbf{C}_{33}$ exactly instead of leaving it to the Neumann series, and leaves the in-plane channel alone. Vertical P incidence excites e_{33} ; vertical S incidence excites e_{13} or e_{23} ; both are corrected. The in-plane shear, which the side-scattered field carries, is unchanged. If the response is dominated by reflection and transmission, the asymmetry is “z treated better” rather than “z treated worse”.

Consequence for any comparison. Do not score the raw coupling against an independent arbiter in a homogeneous background. Three effects are in play at comparable magnitude — the Galerkin-versus-collocation projection, the O_h -invariant resummation, and the artefact — and a single number separates none of them. Any comparison must be channel-resolved.

Whether the resummation is an *improvement* is not established here. The local inversion $1/(K_c + \Delta K_c)$ is exact; whether that makes the closed T -matrix better depends on how the ansatz error and the series truncation interact, and those do not simply add.

8 The propagator, and the bridge to it

8.1 It already exists

Everything downstream needs the Green’s matrix of (1). It does not need to be built: the stratified 6×6 Green’s function and its validated wrapper already return it, in the basis $(u_z, u_x, u_y, T_{zz}, T_{xz}, T_{yz})$ — which is exactly the basis the thesis writes $\mathbf{A}$ in. That the two are the same object had, however, been *assumed*.

8.2 The bridge identity

Away from the source plane the Green’s matrix solves the homogeneous system in its receiver coordinate, so for two receiver depths in one uniform layer

$$\Gamma(z_2, z') = \exp(\mathbf{A}(z_2 - z_1)) \Gamma(z_1, z'), \quad (26)$$

with no free parameters. The identity is *local*: it holds however complicated the medium between the source and z_1 is, because the ODE knows only the layer the receiver sits in. The gate therefore places the source above a fast slab and the receivers below it, so transmission, reflection, conversion and interbed multiples are all present in Γ and (26) must still hold exactly.

8.3 Reporting residuals against the arithmetic floor

A bare relative residual is not comparable across incidence angles. Beyond the critical slowness the propagator carries both a growing and a decaying evanescent branch, so $\exp(\mathbf{A} dz)$ has dynamic range $\sim e^{|k|dz}$ — about 1.6×10^7 at the widest angle used. Forming $\exp(\mathbf{A} dz) \Gamma(z_1)$ then amplifies the rounding already present by $\|\exp(\mathbf{A} dz)\|$, and no correct implementation can beat

$$\text{floor} = \epsilon \|\exp(\mathbf{A} dz)\| \frac{\|\Gamma(z_1)\|}{\|\Gamma(z_2)\|}. \quad (27)$$

Residuals below are quoted in units of that floor, so $O(1)$ means the identity holds to the last bit available at that conditioning.

k_x	k_y	$ k /\omega$	raw	corrected
2.00	1.30	0.0759	7.68×10^{13}	1.26×10^{-2}
0.40	0.05	0.0128	2.95×10^{14}	2.43×10^{-2}
14.00	9.00	0.5298	4.31×10^{13}	9.37×10^{-3}
control, $\zeta \leftrightarrow \chi$ swapped			7.68×10^{13}	1.36×10^{12}

Table 2. Residual of (26) in units of the arithmetic floor. $\omega = 31.4159 \text{ rad s}^{-1}$, $dz = 1 \text{ km}$. The third row is evanescent for both branches.

The corrected object is the physical state vector of $\partial_3 \mathbf{q} = \mathcal{A} \mathbf{q}$; the raw one misses by fourteen orders of magnitude.

The negative control swaps ζ and χ — two in-plane stiffnesses that differ by exactly 2μ , a plausible transcription slip that no symmetry check would catch. It is rejected by fourteen orders.

9 The single-site T -matrix

This is the object the construction was for. On the nine-parameter affine trial space,

$$\boxed{\mathbf{T} = \Delta \mathbf{C}_{\text{eff}} (\mathbf{I}_9 - \mathcal{G} \Delta \mathbf{C}_{\text{eff}})^{-1}}, \quad (28)$$

with $\Delta \mathbf{C}_{\text{eff}}$ the projected coupling of §6 and $\mathcal{G}$ the self propagator moment of the cube.

9.1 What is new and what is borrowed

New is $\Delta \mathbf{C}_{\text{eff}}$. Borrowed is $\mathcal{G}$: the first-order route changes the *coupling*, not the Green's function. The two moments it needs are

$$\mathbf{G}_{ij} = \int_V G_{ij}^0 dV, \quad M_{in,pk} = \int_V \partial'_p \partial'_k G_{in}^0 dV. \quad (29)$$

These are *derived* in §9.2 below, not taken from a table. In the parameter basis

$$\mathcal{G} = \frac{1}{V} \begin{pmatrix} \Gamma_0 \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & -\mathbf{S} \mathbf{W}^{-1} \end{pmatrix}, \quad \mathbf{W} = \text{diag}(1, 1, 1, 2, 2, 2), \quad (30)$$

where $\mathbf{W}$ undoes the engineering doubling of the shear slots.

Note that (30) is asserted, not derived — which is exactly why §9.5 exists.

9.2 The self moments, derived

The integrand of M goes as $1/r^3$, which is not absolutely integrable in three dimensions. The customary remedy is to quote a tabulated Eshelby tensor and add a delta-function term at $r = 0$ by hand; omit the delta and the depolarisation comes out with the *wrong sign*. Nothing below is tabulated and nothing is added by hand.

9.2.1 The static Green's tensor

In the static limit the elastodynamic tensor reduces to Kelvin's,

$$\mathbf{G}_{in}^0(\mathbf{r}) = \frac{a_0 \delta_{in}}{r} + \frac{b_0 x_i x_n}{r^3}, \quad a_0 = \frac{1}{8\pi} \left(\frac{1}{\mu} + \frac{1}{\lambda + 2\mu} \right), \quad b_0 = \frac{1}{8\pi} \left(\frac{1}{\mu} - \frac{1}{\lambda + 2\mu} \right), \quad (31)$$

the coefficients being fixed by requiring that (31) satisfy the Navier operator away from the origin. An equivalent form makes the distributional content visible, since $\partial_i \partial_n r = \delta_{in}/r - x_i x_n/r^3$:

$$G_{in}^0 = \frac{1}{4\pi\mu} \left[\frac{\delta_{in}}{r} + h_c \partial_i \partial_n r \right], \quad h_c = -\frac{\lambda + \mu}{2(\lambda + 2\mu)}, \quad (32)$$

and $a_0 = (1 + h_c)/4\pi\mu$, $b_0 = -h_c/4\pi\mu$ identically.

9.2.2 The moment as a distribution, and why no delta is inserted

Write the moment as a pairing against the test function $w \mathbf{1}_V$ rather than as an integral of a function. With $T = \partial_{\text{rest}} r^m$ and $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$,

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (33)$$

That is the entire construction. Two features make it exact rather than merely plausible:

- T is only ever evaluated *on the boundary*, at distance $\Delta/2$ from the origin, where every derivative of r^m is smooth and bounded. The singular pointwise structure of $\partial^D r^m$ at the origin is never formed, so there is no principal value to take and no delta to insert.
- the recursion strictly lowers both the derivative count and the weight degree, terminating at $\int_V x_w r^m dV$, an ordinary convergent integral for $m \geq -1$.

The delta term is therefore neither added nor droppable — it is already inside the surface integral. The sharp test is the Laplacian sum rule $\sum_p E[-1; \{p, p\}] = -4\pi$, which a spherical excision returns as 0.

9.2.3 The face integrals, and one of them by pure geometry

On the face $x_3 = \Delta/2$, scaling the in-face coordinates by $\Delta/2$ reduces every surface term to one of three integrals over $[-1, 1]^2$:

$$j_1 = \iint \frac{du dv}{(1 + u^2 + v^2)^{3/2}} = \frac{2\pi}{3}, \quad j_2 = \iint \frac{u^2 du dv}{(1 + u^2 + v^2)^{5/2}} = -\frac{2(\sqrt{3} - \pi)}{9}, \quad k_1 = \iint \frac{du dv}{(1 + u^2 + v^2)^{5/2}} = \frac{2(2\sqrt{3} + \pi)}{9}. \quad (34)$$

All three are evaluated in closed form. The first needs no integration at all: j_1 is the solid angle subtended by one face at the cube's centre, and the six faces subtend 4π , so $j_1 = 4\pi/6 = 2\pi/3$. That identity is a check on the scaling and orientation of (34), obtained independently of any quadrature.

The zeroth moment needs one further integral, which also has a closed form:

$$\int_{[-1,1]^3} \frac{dV}{r} = -2(\pi + \log 64 - 12 \log(1 + \sqrt{3})) = \frac{4}{3} \log(70226 + 40545\sqrt{3}) - 2\pi = 9.520309455918214 \dots \quad (35)$$

The two closed forms are independent evaluations of the same constant and agree to the precision carried; the second is the one this package uses elsewhere.

9.2.4 The closed forms

Assembling (31) through the recursion (33) with the face integrals (34) gives, with

$$M_{in,pk} = A \delta_{in} \delta_{pk} + B (\delta_{ip} \delta_{nk} + \delta_{ik} \delta_{np}) + C E_{inpk}, \quad E_{inpk} = \begin{cases} 1 & i = n = p = k \\ 0 & \text{otherwise,} \end{cases} \quad (36)$$

the third structure being the one a cubic — as opposed to isotropic — geometry admits,

$$B = \frac{\sqrt{3}(\lambda + \mu)}{6\pi\mu(\lambda + 2\mu)}, \quad (37)$$

$$A = B - \frac{1}{3\mu}, \quad (38)$$

$$C = -\left(5 - \frac{2\pi}{\sqrt{3}}\right)B, \quad (39)$$

$$G_{ij} = \delta_{ij} \left(a_0 + \frac{b_0}{3}\right) \left(\frac{\Delta}{2}\right)^2 \int_{[-1,1]^3} \frac{dV}{r}. \quad (40)$$

A , B and C are scale-free; G carries Δ^2 . Note that A , B and C are *not* three independent constants — the whole rank-four moment is carried by B together with the elementary $1/3\mu$, which is a structural statement the tabulated presentation obscures. G_{ij} must be a multiple of δ_{ij} , because a rank-two cubic invariant has no other form: the cube cannot distinguish its own axes at that order.

9.2.5 The check that certifies the delta content

From (32) with $\nabla^2(1/r) = -4\pi\delta^3$ and $\nabla^2(\partial_i \partial_n r) = 2\partial_i \partial_n(1/r)$, whose cube integral is $-\frac{4\pi}{3}\delta_{in}$,

$$\sum_p M_{in,pp} = -\frac{\delta_{in}}{\mu} \left(1 + \frac{2h_c}{3}\right) = -\delta_{in} \frac{2\lambda + 5\mu}{3\mu(\lambda + 2\mu)}, \quad (41)$$

derived without reference to the recursion at all. The assembled moment satisfies (41) identically, and the export refuses to write its output otherwise. A spherically excised moment returns zero on the left-hand side, so (41) is exactly the statement that the $r = 0$ content is present and correctly weighted.

9.3 The tabulated route agrees

This package also carries the conventional tabulated Eshelby route, in which j_1 , j_2 and k_1 are quoted rather than integrated. Evaluated alongside (37)–(40) at $\lambda = 17.5$, $\mu = 22.5$ GPa, $\Delta = 1$:

moment	integrated	tabulated	relative
A	$-1.220110745874 \times 10^{-11}$	$-1.220110745874 \times 10^{-11}$	0
B	$2.613707356074 \times 10^{-12}$	$2.613707356074 \times 10^{-12}$	3.1×10^{-16}
C	$-3.587055298887 \times 10^{-12}$	$-3.587055298887 \times 10^{-12}$	2.3×10^{-16}
G	$6.622004020056 \times 10^{-12}$	$6.622004020056 \times 10^{-12}$	0

Table 3. The table reproduces the integration to machine precision. The direction of that statement matters: the integration *verifies* the table; the assembly reads the integrated values.

The dynamic (smooth radiation) part carries no table — it is already exact polynomial integration — and is taken from the existing route with its tabulated static piece removed and the integrated one substituted.

9.4 Two implementations of the coupling

$\Delta\mathbf{C}_{\text{eff}}$ is rebuilt in Python directly from $\Delta\mathcal{A}$, independently of the symbolic derivation, and compared against the values that derivation exports. Worst relative difference over three contrasts: 1.4×10^{-15} for $\Delta\mathbf{C}_{\text{eff}}$ and exactly zero for the Navier coupling. Two implementations agreeing is the evidence standard; one agreeing with itself is not.

9.5 Calibration: every normalisation fixed against existing code

Run the *same* assembly with the ordinary Navier coupling in place of $\Delta\mathbf{C}_{\text{eff}}$. It must reproduce this package’s validated four amplification factors. That pins the volume factors, the sign of $\mathcal{G}$ and the weight $\mathbf{W}$ — the *bookkeeping* of (30) — with no freedom left to tune. It does not certify the moments themselves; that is what the integration of the previous subsection is for. Against $\alpha = 5000$, $\beta = 3000$, $\rho = 2500$ (SI), $\omega = 60$, $a = 0.5$:

case	factor	assembled vs existing (rel.)
pure shear ($\Delta\mu = 1$ GPa)	$A_u, A_\theta, A_e^{\text{diag}}, A_e^{\text{off}}$	0, 2.2×10^{-16} , 0, 0
moderate (2, 1, 100)	same four	0, 2.3×10^{-16} , 1.1×10^{-16} , 0

Table 4. The assembly reproduces the existing amplification factors to machine precision. Only after this is the coupling swapped.

One term had to be isolated to get there. The weak form’s $\int_V \Delta\rho \omega^2 \phi \cdot \mathbf{u} dV$ contributes to the *strain*–strain block as well as the rigid one, because an affine trial displacement is not constant: $\int_V x_a x_b dV = \delta_{ab} V a^2 / 3$. This is a finite-size dynamic coupling between the density contrast and the strain channels, and the factorised amplification factors do not carry it — there A_u and the strain amplifications are independent. It shifts A_θ by 1.5×10^{-7} at the moderate contrast and vanishes identically with $\Delta\rho$. It is a small correction that the factorisation drops, not a defect in either object, and it is reported rather than absorbed.

9.6 The first-order T , channel by channel

A single norm would average a correction that lives in a few channels against others that do not move. The comparison is therefore resolved by channel.

case	channel	relative difference from the Navier route
pure shear	θ (bulk)	1.13×10^{-4}
	e_{diag}	1.1×10^{-16}
	e_{12} (in-plane shear)	0
	e_{13} (out-of-plane shear)	8.36×10^{-4}
	u (density)	0
moderate	θ (bulk)	1.74×10^{-3}
	e_{diag}	1.1×10^{-16}
	e_{12}	0
	e_{13}	8.36×10^{-4}
	u	0

Table 5. Amplification $(\mathbf{I} - \mathcal{G}\Delta\mathbf{C})^{-1}$ along each channel. The density channel is identical in both routes, as §7 requires; the in-plane shear is untouched; the bulk and out-of-plane shear channels move.

9.7 The broken symmetry, now visible in T itself

e_{12} and e_{13} are equivalent under O_h . For an isotropic cube in an isotropic background their amplifications *must* be equal. They are, in the Navier route, to exactly zero. In the first-order route they split:

	Navier	first-order
$ A(e_{12}) - A(e_{13}) / A(e_{12}) $	0	8.36×10^{-4}

Table 6. The O_h violation of §7.4, measured on the T -matrix rather than on the coupling. Identical at both contrasts, because both carry the same $\Delta\mu$ and the shear channels see only that.

This is the sharpest form of the result. It is not a norm over a matrix: it is two numbers that a symmetry argument says must agree, and do not.

What the split does and does not license. It is tempting to read this as the first-order T failing an external test while the ordinary route passes. That reading claims more than the check can deliver, and the two halves must be kept apart.

What it gives. An **arbiter-free lower bound on the error of the first-order T** . If $A(e_{12})$ and $A(e_{13})$ must be equal and differ by δ , then whichever lies closer to the truth, the larger of the two errors is at least $\delta/2 \approx 4.2 \times 10^{-4}$. An absolute error estimate obtained from a symmetry alone, with nothing external to compare against, is worth having — and it is the absolute companion this project requires alongside any ratio test.

What it does not give. Any bound whatever on the ordinary route’s error. A symmetric approximation can be symmetrically wrong, missing the truth in *both* channels by more than 8.4×10^{-4} while satisfying the symmetry exactly. Symmetry compliance is not accuracy. Setting a bound on one formulation against no bound on the other is not a comparison, and no ordering of the two follows from this table.

What is nonetheless attributable. The breaking is not manufactured by the projection mismatch of §9: $\mathcal{G}$ is itself O_h -symmetric, and §7.4 measured the coupling's asymmetry symbolically with no propagator present at all. The asymmetry is a property of the coupling, and it survives into $\mathbf{T}$.

The channel set has to include e_{13} . The coupling leaves e_{12} untouched by construction, so a comparison spanning only θ , e_{diag} and e_{12} reports that the shear channels agree — true, and uninformative. A channel-resolved comparison is only as good as the channels chosen, and the O_h -equivalent pair is the one that carries the information.

9.8 What this does and does not settle

The Born limits agree: at a contrast of 10^{-6} the two couplings differ by 4.2×10^{-8} , i.e. at $O(\Delta c)$ as they must.

The caveat that no number above reveals. The propagator moment used is the *single* (collocation) one, while the $\mathbf{J}_6$ pairing makes the scheme Galerkin. The consistent pairing needs the *double* moment. This is the same class of mismatch the package has already measured as a genuine error elsewhere, so the table above should be read as the first-order coupling evaluated through a collocation propagator, not as the fully consistent first-order T . No winner is declared, and none should be read in.

10 The propagator moment from the first-order system

§9 took $\mathcal{G}$ from the ordinary Kelvin tensor, so the T -matrix assembled there is first-order in its *coupling* only. This section replaces that half. The plan is `docs/plans/2026-09-18-first-order-propagator-moment.md`; what follows is written as each step lands, and says plainly where the front is.

The object is the sandwiched form

$$[\Delta \mathbf{C}_{\text{eff}} \mathcal{G} \Delta \mathbf{C}_{\text{eff}}]_{kl} = \langle \mathbf{J}_6 \psi_k \mid \Delta \mathcal{A} \Gamma \Delta \mathcal{A} \mid \psi_l \rangle. \quad (42)$$

Plain Galerkin on $\mathbf{q}$ is not available: its mass matrix $\langle \mathbf{J}_6 \psi_k, \psi_l \rangle$ is antisymmetric, and an antisymmetric matrix in *odd* dimension — here nine — is singular.

10.1 Step 1: Γ , built from $\mathcal{A}$ alone

For $\partial_3 \mathbf{q} = \mathbf{A} \mathbf{q} + \mathbf{d} \delta(z - z')$ the radiating Green's matrix is

$$\Gamma(z, z') = \begin{cases} +\mathbf{P}_d e^{\mathbf{A}(z-z')}, & z > z', \\ -\mathbf{P}_u e^{\mathbf{A}(z-z')}, & z < z', \end{cases} \quad (43)$$

with $\mathbf{P}_d, \mathbf{P}_u$ the spectral projectors onto the down- and up-going eigenspaces, so that Γ decays away from the source in both directions and jumps by $\mathbf{P}_d + \mathbf{P}_u = \mathbf{I}$ across it.

Projectors, not eigenvectors. The thesis writes the eigen-matrix column by column with normalisation factors $\epsilon_P, \epsilon_S, \epsilon_H$ left symbolic. Those factors are fixed by a flux normalisation that degenerates at normal incidence, where the horizontal direction no longer distinguishes the *SV* and *SH* polarisations — exactly the $k_{\parallel} \rightarrow 0$ difficulty. The spectral projector

$$\mathbf{P}_d = \mathbf{V}_d (\mathbf{W}_d^t \mathbf{V}_d)^{-1} \mathbf{W}_d^t \quad (44)$$

is invariant under any rescaling of the columns of $\mathbf{V}_d$, so building $\mathbf{\Gamma}$ from projectors removes the ambiguity instead of carrying it. That invariance is verified directly, by rescaling every eigenvector by a random complex factor and confirming $\mathbf{P}_d$ is unchanged.

The matrix exponential must never be formed. Equation (43) is a statement, not an algorithm. Evaluating it as written — projecting *after* exponentiating — builds the growing branch and then cancels it. At a separation of 640 km the growing entries reach $e^{+28} \approx 10^{12}$, so the cancellation leaves 10^{-4} of absolute noise on a true answer of 3×10^{-8} . This is the Thomson–Haskell instability, the reason invariant imbedding exists, reappearing inside our own construction; it announces itself as a decay rate that turns *negative* at large separation. Applying the exponential to the projected eigenvalues never forms the growing branch at all,

$$\mathbf{\Gamma}(z - z') = \pm \sum_{i \in \text{down/up}} e^{s_i(z-z')} \mathbf{v}_i \mathbf{w}_i^t, \quad \mathbf{w}_i \cdot \mathbf{v}_j = \delta_{ij}, \quad (45)$$

and is what is used. The moment integral needs only $|z - z'| \leq 2a$, where the instability cannot bite; a layered background with thick layers is where it would.

ω is complex from the outset. The vertical wavenumbers $k_{z,c} = \sqrt{k_c^2 - k^2}$ vanish on the circles $|k| = k_P$ and $|k| = k_S$, which the lateral integral must cross, and k_z sits in a denominator. At real ω a quadrature straddles an integrable $1/\sqrt{\cdot}$ singularity and does not converge; the directional-sweep work measured $1.9 \times 10^{-1} \rightarrow 1.1 \times 10^{-1}$ over an eightfold refinement, non-monotone. With damping 0.02 the smallest $|s|$ on the P circle is 0.240 rather than zero.

Checks. Thirteen, all passing: the spectrum is $\{\pm i k_{z,P}\}$ and $\{\pm i k_{z,S}\}$ twice; $\mathbf{P}_d + \mathbf{P}_u = \mathbf{I}$, both idempotent, both commuting with $\mathbf{A}$; $\mathbf{P}_d$ invariant under rescaling; $\partial_z \mathbf{\Gamma} = \mathbf{A} \mathbf{\Gamma}$ on both sides; the jump exactly $\mathbf{I}$; and the decay rate converging to the slowest eigenvalue, $0.034990 \rightarrow 0.033901 \rightarrow 0.032857 \rightarrow 0.032436$ against 0.032393 — a rate, not a bar, because at any one finite separation the faster mode still contributes.

10.2 Step 2: the vertical integral, in closed form

The vertical integral is exact, but it is *not* an integral over the square. The two halves of $\mathbf{\Gamma}$ carry different matrices, so the double integral splits over the two triangles and they cannot be recombined: treating the kernel as a function of $|z - z'|$ alone would symmetrise the propagator and destroy the up/down distinction the first-order system exists to carry.

With affine trial fields the weights are 1 and z , so four triangular integrals are needed,

$$T_{mn}(q) = \int_{-a}^a dz \int_{-a}^z dz' z^m z'^n e^{-q(z-z')}, \quad (46)$$

the lower triangle being T_{nm} by relabelling. All four are elementary:

$$T_{00} = \frac{2aq - 1 + e^{-2aq}}{q^2}, \quad (47)$$

$$T_{10} = -T_{01} = \frac{aq - 1 + (1 + aq)e^{-2aq}}{q^3}, \quad (48)$$

$$T_{11} = \frac{3 - 3(1 + aq)^2 e^{-2aq} + a^2 q^2 (2aq - 3)}{3q^4}. \quad (49)$$

Summing the two triangles with a common matrix recovers the familiar square integral $2(2aq - 1 + e^{-2aq})/q^2$, and the odd weight cancels between them by parity — both checked, as is the large- q behaviour $T_{00} \sim 2a/q$, each triangle carrying half of the square's $4a/q$.

The decisive check of this step is not the closed forms themselves but the *assembly*: the vertical integration of Γ performed through (47)–(49), matrices and all, reproduces a brute-force double quadrature of Γ to 2.5×10^{-9} relative.

10.3 Step 3: the lateral integral

Count powers as $|k| \rightarrow \infty$ for the *untransferred* form, in which $\partial_\alpha \rightarrow ik_\alpha$: the box form factor gives k^{-2} , (47) gives $q^{-1} \sim |k|^{-1}$, the two contrast operators give k^2 , and the measure gives k . The strain channel does not converge. (The transferred form of §10.3.6 does; that is one of the things the augmentation buys, and it is taken up there.)

The divergence is not an artefact to be trimmed. A delta function in real space is a constant in k -space, so the non-decaying tail of the integrand *is* the Eshelby term of §9.2. A bare cutoff discards precisely the piece whose omission reverses the sign of the depolarisation. Hence split-and-subtract,

$$I(\mathbf{k}) = [I(\mathbf{k}) - I_\infty(\mathbf{k})] + I_\infty(\mathbf{k}), \quad (50)$$

with the bracket integrated numerically — convergent, and its convergence measured as a rate — and $\int I_\infty d^2k$ evaluated in closed form over the whole plane, where the delta content resides. The asymptote is taken of the z -integrated kernel (47)–(49), with the box form factors kept exact: they are what makes the remainder converge, so expanding them would defeat the subtraction.

10.3.1 The scaling that makes the expansion uniform

$\mathbf{A}$ is not uniformly $O(k)$. Rows 1–3 carry k , but rows 5–6 carry k^2 , because the basis mixes displacement with traction and the two have different dimensions; expanding $\mathbf{A}$ in $1/k$ as it stands mixes orders between blocks. The traction of an evanescent field scales as $\mu k u$, so the natural balance is $\mathbf{q} \rightarrow (\mathbf{u}, \boldsymbol{\tau}_3/k)$. Under $\mathbf{S} = \text{diag}(1, 1, 1, k, k, k)$,

$$\tilde{\mathbf{A}} = \mathbf{S}^{-1} \mathbf{A} \mathbf{S} \quad (51)$$

is uniformly $O(k)$: the upper-right block gains a factor k , the lower-left loses one, and the diagonal blocks are untouched. Measured, $\|\mathbf{A}\|/k$ grows as $5.3 \times 10^3 \rightarrow 5.3 \times 10^5$ over two decades in k while $\|\tilde{\mathbf{A}}\|/k$ settles at $53.0100 \rightarrow 53.0101$. Being a similarity transform it changes no eigenvalue, and $|s|/k \rightarrow 1$ for all six.

The scaling is also what makes the decomposition numerically usable. On the unscaled matrix the down- and up-going eigenvectors become nearly parallel as k grows — the evanescent limit degenerates — the eigen-decomposition is reported badly conditioned, and the kernel entries collapse into rounding error, from which no asymptotic power can be read. The decomposition is therefore done on $\tilde{\mathbf{A}}$ and mapped back, $\mathbf{K} = \mathbf{S} \tilde{\mathbf{K}} \mathbf{S}^{-1}$, checked against the verified kernel of step 2 at moderate k .

10.3.2 Which entries fail to decay

The order in $1/k$ that each entry of the z -integrated kernel carries is read off numerically, over decades in k , rather than derived: a measured power is harder to fool oneself with than one obtained from a 6×6 eigen-perturbation. A dot marks an entry that vanishes, so no power can be read.

	u_z	u_x	u_y	τ_{zz}	τ_{xz}	τ_{yz}
u_z	.	-1	-1	-2	.	.
u_x	-1	.	.	.	-2	-2
u_y	-1	.	.	.	-2	-2
τ_{zz}	0	.	.	.	-1	-1
τ_{xz}	.	0	0	-1	.	.
τ_{yz}	.	0	0	-1	.	.

Table 7. Power of k carried by each entry of the z -integrated kernel as $|k| \rightarrow \infty$. Exactly one block fails to decay.

The entire ultraviolet problem sits in one block: the traction rows against the displacement columns. Everything else decays at least as $1/k$ and needs no subtraction.

That block is also the one the physics predicts. In the evanescent limit a short-wavelength displacement produces a traction $\tau \sim \mu k u$, while the vertical integral (47) supplies $1/k$; the ratio is k -independent. A traction response to a displacement that does not depend on k is a local stiffness — which is the Eshelby delta of §9.2, seen from the first-order side. The two representations identify the same object, and I_∞ has to reproduce exactly this block.

10.3.3 The asymptote in closed form

The limit of the non-decaying block is not merely numerical. Writing $\hat{\mathbf{k}}$ for the lateral direction,

$$C[\tau_{zz} \leftarrow u_z] = -v_1 = -\frac{4\mu(\lambda + \mu)}{\lambda + 2\mu}, \quad (52)$$

$$C[\tau_{\alpha z} \leftarrow u_\beta] = \mu(\hat{k}_\alpha \hat{k}_\beta - \delta_{\alpha\beta}). \quad (53)$$

ν_1 **is not a new constant.** It is the same ν_1 that appears in $\mathcal{A}_{12}$ — born in the elimination (7), and shown in §3.2 to be the thesis’s ζ . The subtraction is expressible entirely in the operator’s own constants. A wrongly identified asymptote would have no reason to land on ν_1 and μ .

Both are established as *rates*. The relative residuals fall $1.50 \times 10^{-3} \rightarrow 1.50 \times 10^{-4} \rightarrow 1.50 \times 10^{-5}$ across decades in k , ratios 10.0006 and 10.011 — clean $O(1/k)$, which is the $O(k_c/k)$ correction to the evanescent limit. Agreement at a single k would not distinguish a correct closed form from a coincidence.

The in-plane form is the sharper test. Two constants are fitted on *two* angles and then predict *six*: $P = -22.4977$ and $Q = +22.4948$, both equal to $\mu = 22.5$, with predictions tracking measurements to five digits across $\theta \in [0, 2.4]$. The isotropic entry (52) is constant to seven digits over the same range.

What this settles about the integration. The asymptote *is* direction-dependent — the relative variation over θ is 0.39 — so I_∞ is not a pure delta and the regulator cannot be avoided. But the structure is the most favourable available: $\delta_{\alpha\beta}$ and $\hat{k}_\alpha \hat{k}_\beta$ are the only two rank-two invariants a direction can build, and they transform to a local delta and a principal-value dipole kernel respectively, both standard. The $\tau_{zz} \leftarrow u_z$ entry is isotropic and so is a pure delta of weight $-\nu_1$.

10.3.4 The lateral integral, and where it must be evaluated

The box form factor is separable — the lateral transform of the cube’s indicator is $(2a)^2 \text{sinc}(k_1 a) \text{sinc}(k_2 a)$ — so with (52)–(53) and one lateral derivative from each $\Delta\mathcal{A}$, the integrand carries $k_c k_d$ against C , giving two families: $\int d^2 k W k_c k_d$, which is separable, and $\int d^2 k W k_a k_b k_c k_d / k^2$, which is not. The elementary one-dimensional integrals are

$$\int_{-\infty}^{\infty} \text{sinc}^2(ka) dk = \frac{\pi}{a}, \quad \int_{-\infty}^{\infty} k^2 \text{sinc}^2(ka) e^{-\epsilon|k|} dk = \frac{4}{4a^2\epsilon + \epsilon^3} = \frac{1}{a^2\epsilon} - \frac{\epsilon}{4a^4}. \quad (54)$$

The second is a pure pole with no finite part. So $\int I_\infty d^2 k$ is not a number, and the split-and-subtract cannot be closed in k -space alone: adding back a pure pole restores exactly the divergence it was introduced to remove.

The method is not wrong; the tail is being evaluated in the wrong representation. Forming $|\hat{S}|^2$ first and then integrating term by term destroys the structure that makes the constant part finite. By Parseval,

$$\iint f(\mathbf{x}) K(\mathbf{x} - \mathbf{x}') g(\mathbf{x}') = \int \frac{d^2 k}{(2\pi)^2} \hat{f}(-\mathbf{k}) \hat{K}(\mathbf{k}) \hat{g}(\mathbf{k}), \quad (55)$$

so a $\hat{K}$ tending to a constant C contributes $C \int_V f g d\mathbf{x}$ — a *local* term, finite, needing no regulator at all. This is verified directly in one dimension, both sides returning $1/2a$.

The decomposition therefore splits by representation rather than by magnitude:

piece	real-space object	evaluation
isotropic, $-\nu_1$ and $P\delta_{\alpha\beta}$	delta	local, elementary
traceless $\hat{k}_\alpha\hat{k}_\beta - \frac{1}{2}\delta_{\alpha\beta}$	2-D principal-value dipole	conditionally convergent
remainder $I - I_\infty$	—	numerically, in k

Table 8. Split-and-subtract survives, but $\int I_\infty$ is evaluated in real space. Only the remainder stays in k .

The traceless part is of degree -2 with vanishing angular average — exactly the conditional convergence the real-space engine of §9.2 handles by peeling derivatives onto the cube faces. The two representations meet again here: the same distributional structure, in two dimensions rather than three.

10.3.5 The principal-value piece vanishes

In two dimensions $\nabla^2 \log \rho = 2\pi\delta^2$, and $\partial_\alpha \partial_\beta \log \rho = (\delta_{\alpha\beta} - 2\hat{n}_\alpha \hat{n}_\beta)/\rho^2$ away from the origin; taking the trace fixes the delta weight, so that

$$\partial_\alpha \partial_\beta \log \rho = \text{PV} \left[\frac{\delta_{\alpha\beta} - 2\hat{n}_\alpha \hat{n}_\beta}{\rho^2} \right] + \pi \delta_{\alpha\beta} \delta^2(\mathbf{x}). \quad (56)$$

By translation invariance the double integral over the square collapses onto the autocorrelation of its indicator, $w(\mathbf{u}) = (2a - |u_1|)(2a - |u_2|)$, exactly — removing two of the four dimensions — and both derivatives move onto w , which is legitimate because w has compact support.

w is piecewise bilinear, so $\partial_1 \partial_1 w$ is a sum of line deltas. It has three terms, not one: w vanishes at $|u_1| = 2a$ but $\partial_1 w$ does *not* — it jumps from $\mp(2a - |u_2|)$ to zero — so there are edge deltas at $u_1 = \pm 2a$ alongside the interior one at $u_1 = 0$. With them,

$$D_{11} = \underbrace{3}_{\text{interior}} + \underbrace{(\pi - 3)}_{\text{edges}} = \pi = \pi(2a)^2 \Big|_{a=1/2}, \quad D_{12} = 0, \quad (57)$$

and the trace identity $\text{tr } D = \langle \nabla^2 \log \rho, w \rangle = 2\pi w(0) = 2\pi(2a)^2$ holds exactly. Dropping the edge deltas gives 6 in place of 2π , so that identity is what detects them.

Hence $\langle \text{PV}_{\alpha\beta}, w \rangle = D_{\alpha\beta} - \pi \delta_{\alpha\beta} w(0) = 0$: the traceless dipole contributes nothing.

It could not have contributed. $\langle \text{PV}_{\alpha\beta}, w \rangle$ is a symmetric two-tensor invariant under the square's D_4 symmetry, and the only such tensor is a multiple of $\delta_{\alpha\beta}$ — the traceless part has nowhere to live. The conditional convergence is real, but for this cross-section it integrates away.

The whole of $\int I_\infty$ is therefore the local delta term. The piece that looked hardest — the conditionally convergent, shape-dependent dipole — is identically zero here, and what remains is elementary.

10.3.6 Why the transferred form converges

In the untransferred form $\partial_\alpha \rightarrow ik_\alpha$, each contrast operator contributes a factor of k , and the pole of (54) follows. After the by-parts transfer of (17) the derivatives act on the *polynomials, in real*

space: ∂_α of an affine trial function is a constant, whose lateral transform is again the box form factor. No factor of k survives, and the lateral integral converges.

The augmentation therefore does two things at once: it keeps the indicator from being differentiated, and it makes the lateral integral convergent. Both are the same integration by parts.

Convergence is nevertheless slow. With $\hat{K} \rightarrow C$ the truncation error falls only as $1/\Lambda$, which is what the subtraction is for.

10.3.7 The tail, in closed form

The tail is elementary. For the in-plane entry $C = \mu(\hat{k}_1^2 - 1) = -\mu k_2^2/k^2$, so

$$\int d^2k W C = -\mu \int d^2k \operatorname{sinc}^2(k_1 a) \operatorname{sinc}^2(k_2 a) \frac{k_2^2}{k^2} = -\frac{\mu}{2} \left(\frac{\pi}{a}\right)^2, \quad (58)$$

because $k_1^2/k^2 + k_2^2/k^2 = 1$ and the weight is symmetric under $k_1 \leftrightarrow k_2$, so each piece is exactly half of $\int d^2k W = (\pi/a)^2$. No quadrature is required; direct integration returns 19.7376 against $(\pi/a)^2/2 = 19.7392$, the difference being the quadrature's own error on a slowly converging integrand.

10.3.8 The remainder, and the test that closes the construction

Truncating the lateral integral at $|k| < \Lambda$ for the $\tau_{xz} \leftarrow u_x$ entry, with and without the subtraction:

The split is only worth making if both halves give the same answer. Truncating at $|k| < \Lambda$ for the $\tau_{xz} \leftarrow u_x$ entry:

Λ	unsubtracted	remainder	remainder + tail
80	-318.2214	118.83504	-325.29716
160	-321.6764	118.82727	-325.30493
320	-323.4539	118.82513	-325.30707
640	-324.3400	118.82466	-325.30754

Table 9. The subtracted route is converged to six figures by $\Lambda = 80$; the unsubtracted one is still 0.97 short of the limit at $\Lambda = 640$.

The remainder's successive differences fall $7.8 \times 10^{-3} \rightarrow 2.1 \times 10^{-3} \rightarrow 4.7 \times 10^{-4}$, a ratio near 0.27, against the unsubtracted 0.50 — the latter being the $1/\Lambda$ decay the constant part imposes. Extrapolating both:

$$\underbrace{-325.226}_{\text{unsubtracted limit}} \quad \text{versus} \quad \underbrace{-325.308}_{\text{remainder+closed-form tail}}, \quad \text{relative difference } 2.5 \times 10^{-4}, \quad (59)$$

and that residue is dominated by the single-term extrapolation of the slowly converging unsubtracted series, not by the construction. At $\Lambda = 640$ the unsubtracted error is 0.97 and the subtracted one 5×10^{-4} : a factor of about two thousand.

Split-and-subtract is verified. The two routes to the same number agree, the tail is closed in form by (58), and the remainder converges three orders of magnitude faster than the direct integral.

Status of Task 1. The method is established end to end: Γ from spectral projectors; the triangular z integrals and their assembly; the scaling, the power table, and the asymptote $-\nu_1$, $\mu(\hat{k}\hat{k} - \delta)$ verified as an $O(1/k)$ rate; $\int I_\infty$ decomposed by representation, its principal-value part vanishing by D_4 symmetry and its remaining part closed by (58); and the closure test (59). What remains is bookkeeping rather than method: assembling all nine trial functions and the full contrast operator, and reading the shear moment B off the result for comparison with $\sqrt{3}(\lambda + \mu)/6\pi\mu(\lambda + 2\mu)$.

11 Summary of checks

object	what is established	checks
$\mathcal{A}$	built from c_{ijkl} ; matches the published isotropic blocks	4
	matches the thesis $\mathbf{A}(k_x, k_y)$ under one similarity transform	3
	symmetry relations, spectrum, normal incidence	6
	stiffness matrices, $\mathbf{U}_{\alpha l}$, ν_1 , ν_2	9
	source vector, negative control	3
$\Delta\mathcal{A}$	exact contrast operator; Navier body force on a generic field	4
	nonlinearity in the contrast, quantified	3
	augmentation: constant, reproduces the form, divergence exhibited	4
	O_h /trial-space structure, density channel	6
$\Delta\mathbf{C}_{\text{eff}}$	$\mathbf{J}_6\Delta\mathcal{A}$ symmetric; test space derived; controls	5
	9×9 symmetric, rank 9, rigid block, basis trap	7
	Born agreement, $O(\Delta c^2)$ difference, localisation	9
	O_h decomposition, positive and negative controls	6
Γ	propagates by $\exp(\mathbf{A} dz)$ with all multiples present	3
moments	face integrals evaluated; j_1 also by solid angle	4
	$\int dV/r$ in two independent closed forms	1
	B , and $A = B - 1/3\mu$, $C = -(5 - 2\pi/\sqrt{3})B$, G	4
	Kelvin coefficients against the h_c form	1
	the trace identity (41), derived without the recursion	1
	the tabulated route reproduces all four	1
$\mathbf{T}$	coupling rebuilt independently, matches the symbolic derivation	2
	calibration: reproduces the existing amplification factors	1
	the finite-size density-strain term, isolated and sized	2
	density channel identical; O_h split on the e_{12}/e_{13} pair	6
	Born limits agree	1

Table 10. All checks pass. The symbolic scripts report 25/25 and 44/44, the T -matrix assembly 12/12, and the propagator gate exits clean.

12 What is not established

In keeping with the project’s standard that an unvalidated formula be labelled as such, the following are *open*:

1. **The T -matrix uses the collocation propagator, not the consistent one.** $\mathbf{T}$ is assembled in §9 and calibrated exactly, but $\mathcal{S}$ there is the *single* cube moment while the $\mathbf{J}_6$ pairing makes the scheme Galerkin. The consistent pairing needs the *double* moment, which has not been extracted. Everything in §9 is therefore the first-order coupling seen through a collocation propagator.
2. **No comparison against an independent arbiter has been run.** The quantities in §7 and §9 compare two *formulations* with each other, not either against truth. The O_h split is the one partial exception, and it is one-sided: it lower-bounds the first-order error at $\approx 4.2 \times 10^{-4}$ and bounds the ordinary route not at all. **Which formulation is more accurate remains open**, and nothing in this note decides it.
3. **Whether the resummation improves accuracy is unknown.** It is exact as a local inversion; that is not the same claim.
4. O_h **invariance was tested on a subgroup of order 3**, not the full group.
5. **The derivative count for a collocation scheme differs from the Galerkin one.** In the Galerkin pairing the transferred derivative lands on polynomials, so the third-derivative moment cannot arise; under collocation it transfers onto $\mathbf{\Gamma}$ and can. Do not quote the Galerkin count for a collocation build.

13 Where the route belongs

The x_3 preference of §7.4 is an artefact only because the problem posed there — an isotropic cube in an isotropic whole space — has no preferred direction. A layered background does. There the formulation’s directional structure matches the physics rather than contradicting it, the O_h argument does not apply, and the propagator is already available and already bridged to $\mathcal{A}$ by (26).

That is also the case in which the ordinary route has no competitor: the cube moment method rests on volume moments of the whole-space Green’s tensor, and in a stratified background that object does not exist. The first-order form supplies what the moment method cannot.